Heron: Phishing URL Detection

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Overview

Heron is a machine learning project that classifies a URL as **phishing** or **legitimate**. It extracts lexical and host-based features from the URL text (length, special characters, IP-address hosts, suspicious keywords, risky TLDs, entropy and more), trains several classifiers, and uses the best one to score new URLs. It never visits the suspicious site, so analysis is fast and safe.

Project Description

Phishing attacks trick users into entering credentials on fake websites. Many fake URLs share recognisable patterns, such as long hostnames, `@` symbols, raw IP addresses, many hyphens or subdomains, and words like `login` or `verify`. Heron learns these patterns from labelled examples and outputs a phishing probability and a verdict for any URL.

Dataset

- **Kaggle: Phishing Site URLs** (columns `URL`, `Label` with `good` / `bad`): https://www.kaggle.com/datasets/taruntiwarihp/phishing-site-urls
- Download the CSV, rename it `phishing_site_urls.csv` and place it in the same folder as the notebook (or edit `DATA_PATH` in the notebook).
- Any CSV with a URL column and a label column (`1` / `0`, `phishing` / `legitimate`, `bad` / `good`) also works.
- If no dataset file is found, the notebook builds a small synthetic demo set so it still runs. Use the real dataset for final results.

Technologies Used

Python 3.10+, Jupyter Notebook, pandas, NumPy, scikit-learn (Logistic Regression, Random Forest, Gradient Boosting), matplotlib, seaborn, joblib.

Files

File	Purpose
`Pranitha_U_P_Heron.ipynb`	Complete project code
`requirements.txt`	Python dependencies
`Pranitha_U_P_ProjectReport.docx`	Full project documentation
`README.md`	This overview

Setup and Run

```
# 1. (Optional) create a virtual environment
python -m venv venv
source venv/bin/activate        # Windows: venv\Scripts\activate

# 2. Install dependencies
pip install -r requirements.txt

# 3. Place phishing_site_urls.csv next to the notebook, then launch
jupyter notebook Pranitha_U_P_Heron.ipynb
```

Run all cells from top to bottom (**Kernel > Restart & Run All**). The notebook saves the trained model as `heron_model.joblib`.

Using the Model

The last section of the notebook defines `heron_predict(url)`:

```
heron_predict("http://paypal-verify.secure-login-482.xyz/login/index.php")
# {'url': ..., 'phishing_probability': 0.99, 'verdict': 'PHISHING'}
```

Key Information

- Approach: URL-only lexical features (about 30), stratified 80/20 split, 5-fold cross-validation.
- Metrics reported: accuracy, precision, recall, F1, ROC-AUC, confusion matrix.
- Limitations: no page-content, WHOIS or reputation features; may lag behind new phishing tactics.
- Future work: domain-age and DNS features, deep learning on raw URL characters, a web or browser-extension front end.